

UNIVERSITY OF THE PEOPLE

“Project Proposal and Planning: St. Luke Foundation Digital Workplace Platform (SL-DWP)”

**Course: MSIT 5910-01 MSIT Capstone Project - AY2027-
T1**

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I. Project Charter

Charter Element	Description
Project Title	St. Luke Foundation Digital Workplace Platform (SL-DWP)
Project type	Applied Information Technology Capstone
Project domain	Digital workplace / Information systems / Web application
Methodological approach	Design science-oriented approach within a case-study context
Duration	8 weeks
Primary deliverable	Secure Digital Workplace Minimum Viable Product (MVP)
Technology	Laravel/PHP, relational database, HTML/CSS/JavaScript
Security model	Authentication, RBAC, least privilege, and audit logging
Evaluation	Functional, security, usability, performance, and feasibility
Budget Estimate	\$ 11,627.00 CAD
Primary organization	St. Luke Foundation for Haiti
Project status	Proposed

Project Purpose:

The proposed St. Luke Foundation Digital Workplace Platform (SL-DWP) is an eight-week information technology project designed to address information-access, communication, document-sharing, and access-control challenges within a geographically distributed, multi-sector nonprofit organization. The project will produce and evaluate a secure web-based minimum viable product (MVP), rather than attempt to implement a complete enterprise information system. This proposal defines the problem, project scope, SMART objectives, expected outcomes, ethical and security considerations, and an initial project plan.

II. Problem Statement and Organizational Context

1. St Luke Foundation’s Context

St. Luke Foundation operates across approximately 45 sites, including schools, health facilities, a production center, and administrative operations. Its geographically distributed structure creates challenges for maintaining consistent access to organizational information and communication. Periodic disruptions to physical mobility can further increase dependence on reliable digital channels for approved information sharing and coordination.

The problem is primarily situated within information systems, software engineering, cybersecurity, database management, and digital workplace technology. Research on nonprofit organizations indicates that strategic IT alignment can support organizational agility during changing or crisis conditions (Azevedo et al., 2024). Similarly, Almadani et al. (2024) identify integration, centralized information sharing, privacy, security, and performance as important considerations when technology is adopted in nonprofit environments.

The proposed project therefore focuses on a specific IT problem: the absence of a centralized, role-based digital workplace prototype that can provide authorized users with consistent access to selected organizational information and communication resources.

2. Stakeholder Analysis

Stakeholder	Interest / Needs	Influence	Project relevance
Foundation leadership	Strategic information management	High	Defines organizational context
Managers	Communication and coordination	High	Validate workflows

Employees	Information and document access	Medium	Primary users
System administrators	Users, roles, security	High	Validate administration/security
Sector/location representatives	Information availability and secure accessibility	Medium	Validate representative workflows
Test participants	Usability	Medium	Evaluate MVP
Academic supervisor	Academic quality	High	Project oversight
Researcher/developer	Project delivery	High	Design, development, evaluation

3. Problem Statement

St. Luke Foundation for Haiti currently lacks an integrated, secure, role-based digital workplace for centralized organizational information, internal communication, controlled document sharing, and selected administrative coordination across its geographically distributed operations. The proposed project will design, implement, and evaluate an eight-week web-based MVP that addresses these functions while applying role-based access control, basic audit logging, and security-by-design practices.

4. Research Questions

➤ Primary research question

How can a secure, role-based digital workplace MVP be designed and evaluated to support information access, internal communication, controlled document management, and selected coordination within the St. Luke Foundation for Haiti?

➤ Secondary questions

RQ1: What functional, usability, security, and performance requirements are necessary for an MVP digital workplace in the St. Luke organizational context?

RQ2: How effectively can RBAC and least-privilege principles control access to selected organizational resources?

RQ3: To what extent does the developed MVP satisfy predefined functional, security, usability, and performance criteria?

RQ4: What technical, operational, and security limitations should be considered before potential future expansion of the platform?

III. Project Scope, Goals, and Expected Outcomes

1. Project Scope

The project will deliberately limit functionality to an achievable MVP.

➤ In-Scope

The MVP includes:

- ✓ Authentication and session management
- ✓ User management
- ✓ RBAC
- ✓ Organizational dashboard
- ✓ Employee directory
- ✓ Internal announcements
- ✓ Controlled document repository
- ✓ Audit logging

- ✓ Functional testing
- ✓ Security testing
- ✓ Usability evaluation
- ✓ Performance evaluation

➤ **Out-of-Scope**

- ✗ Medical records
- ✗ Accounting
- ✗ Payroll
- ✗ Complete HR management
- ✗ Complete school management
- ✗ Complete hospital management
- ✗ Advanced inventory
- ✗ E-commerce
- ✗ Native mobile applications
- ✗ Advanced AI
- ✗ Full enterprise project management
- ✗ Full ERP functionality
- ✗ Direct Microsoft 365 integration
- ✗ Production-scale deployment

The principal assumptions are that representative stakeholders can provide requirements and feedback, development hardware and software are available, and synthetic or anonymized data can be used. Major constraints include the eight-week academic schedule, limited development resources, stakeholder availability, and the need to avoid handling sensitive operational or medical information.

2. SMART Objectives for the SL-DWP MVP

Objective	Measure	Target / Deadline
1. Requirements: Establish the SL-DWP MVP requirements.	Requirements, use cases, and RBAC matrix completed.	100% of MVP requirements documented and reviewed by Week 2.
2. System Design: Design the SL-DWP architecture and interfaces.	Architecture, database, RBAC, security, and UI/UX prototypes completed.	100% of the seven core components designed by Week 3.
3. MVP Development: Develop and integrate the seven core SL-DWP components.	Components implemented and unit/integration tested.	100% of core components implemented and $\geq 90\%$ of tests passed by Week 6.
4. MVP Evaluation: Evaluate functionality, security, usability, and performance.	Test results and user-task completion rate.	$\geq 95\%$ functional tests passed, 100% unauthorized-access tests blocked, and $\geq 80\%$ user tasks completed by Week 7.
5. Finalization: Resolve major defects and prepare the MVP for deployment and presentation.	Defect resolution, documentation, deployment-readiness review, and presentation.	100% critical and 90% high-priority defects resolved or documented by Week 8.

3. Expected Outcomes

The principal tangible outcome will be a functional SL-DWP MVP. Supporting deliverables will include requirements and use cases, an RBAC authorization matrix, database and system

architecture, security controls, UI/UX interfaces, test results, a feasibility assessment, and final technical documentation. Success will be determined from documented evaluation results rather than assumed improvements to the organization's performance.

IV. Ethical and Security Implications

Because the platform could eventually process organizational information, privacy and access control must be considered from the planning stage. Ethical issues include privacy, informed participation, data minimization, confidentiality, intellectual property, and responsible use of organizational information. User testing should use informed participation and avoid unnecessary collection of personal information. Synthetic or anonymized records will be preferred during development and testing.

Security risks include unauthorized access, excessive privileges, compromised credentials, unauthorized document access, insecure application inputs, and insufficient auditability. The project will therefore apply security-by-design principles.

The NIST Cybersecurity Framework (CSF) 2.0 provides a useful planning reference because it is intended to help organizations understand, assess, prioritize, and communicate cybersecurity risks across different organizational contexts (Pascoe et al., 2024).

Risk/Issue	Planned Mitigation
Unauthorized access	Authentication and server-side authorization
Excessive privileges	RBAC and least-privilege design
Credential compromise	Secure password hashing and session controls
Unauthorized document access	Role/permission checks before access
Data exposure	Data minimization and anonymized test data

Application vulnerabilities	Input validation and security testing
Poor accountability	Basic audit logging
Privacy during testing	Consent and limited data collection
Intellectual-property issues	Use licensed/open-source components appropriately

The project will emphasize the **Govern, Identify, Protect, Detect, Respond, and Recover** perspective of CSF 2.0 while recognizing that an eight-week student prototype cannot implement a complete enterprise cybersecurity program.

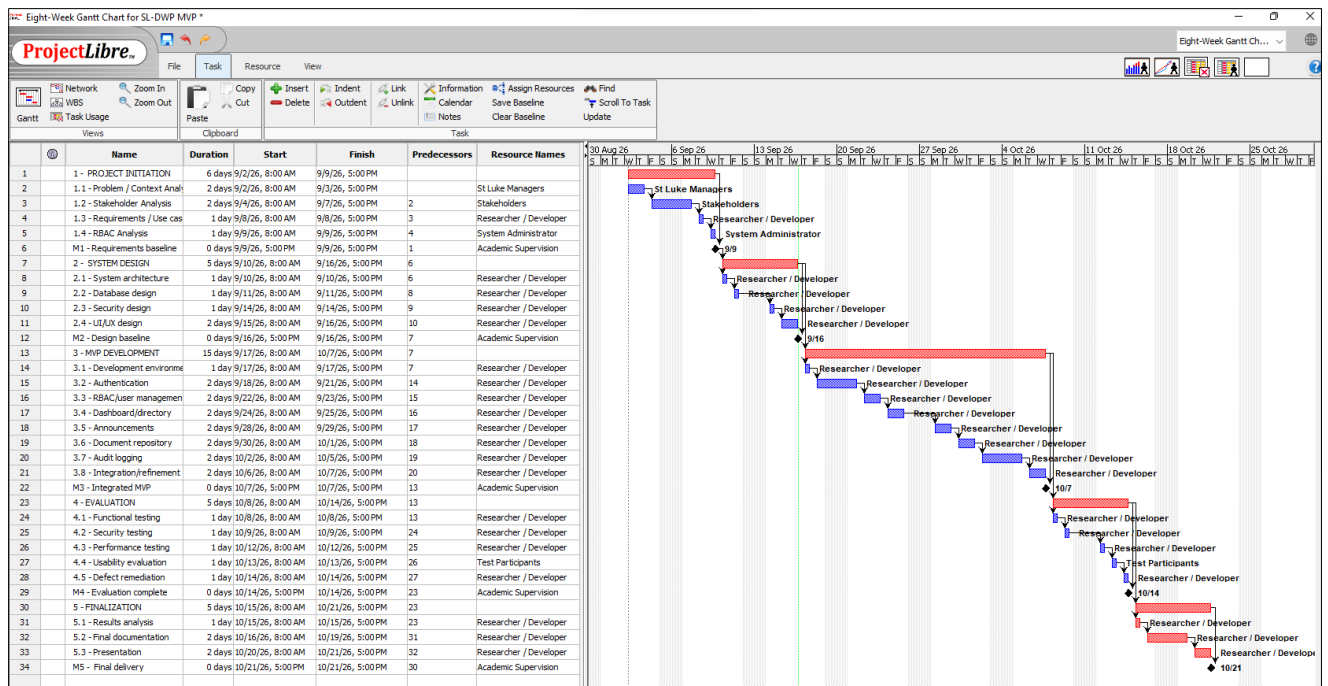
V. Project Planning Tools

1. Work Breakdown Structure (WBS)

The WBS decomposes the project into five major phases and establishes a logical relationship between requirements, design, development, evaluation, and finalization.

ID	Work Package	Major Activities	Deliverable
1.0	Project Initiation	Problem, stakeholders, requirements, RBAC analysis	Requirements baseline
2.0	System Design	Use cases, architecture, database, security, UI/UX	Design baseline
3.0	MVP Development	Environment, authentication, RBAC, modules, integration	Integrated MVP
4.0	Evaluation	Functional, security, performance, usability testing	Evaluation results
5.0	Finalization	Analysis, documentation, presentation	Final project package

2. Eight-Week Gantt Chart



The principal critical-path sequence is requirements, design, MVP development/integration, Evaluation, remediation, finalization. Some design and preparation activities may overlap, but the project cannot proceed to meaningful evaluation until the integrated MVP is available.

3. Resource Justification Table

Resource	Purpose	Justification
Laravel/PHP	Backend/web application	Established framework suitable for rapid MVP development
Relational database	Users, roles, documents, audit data	Structured storage and relational authorization model
HTML/CSS/JavaScript	User interface	Required for browser-based interaction
Git/GitHub	Version control	Tracks changes and supports

		rollback
ProjectLibre	Project planning	Supports WBS, dependencies, milestones, and Gantt planning
Browser developer tools	Testing/debugging	Supports functional and interface testing
Security testing tools	Security evaluation	Helps identify application weaknesses
Synthetic/anonymized dataset	Testing	Provides realistic testing without unnecessary personal data
Existing development computer/server	Development/testing	Avoids unnecessary infrastructure expenditure

Open-source and existing resources are intentionally preferred to maintain economic feasibility.

4. Resource Loading

The following represents a planning estimate, not a claim of actual hours already worked.

Resource	W1	W2	W3	W4	W5	W6	W7	W8	Total
Researcher/developer	25	25	25	25	25	25	25	25	200
Academic supervision	2	2	3	4	2	2	2	3	20
Sector Representatives	16	0	0	0	0	2	6	4	28
St Luke Managers	10	0	0	0	0	0	5	0	15
Test participants	0	0	0	0	0	0	20	0	20
System Administrator	6	0	0	0	0	0	0	0	6
Total hours	59	27	28	29	27	29	58	32	289

The highest resource demand occurs during development, integration, and evaluation, which is consistent with the project's technical workload.

5. Risk Register

ID	Risk	Prob.	Impact	Score	Mitigation	Contingency
R01	Scope expansion	High	High	Critical	Freeze MVP scope	Defer non-core features
R02	Development delay	Medium	High	High	Prioritize core modules	Reduce optional functionality
R03	Stakeholder unavailability	Medium	High	High	Collect requirements early	Use validated representative assumptions
R04	Security vulnerability	Medium	High	High	Security-by-design/testing	Disable affected feature and remediate
R05	Requirements ambiguity	Medium	High	High	Requirements baseline	Revalidate requirements
R06	Integration problems	Medium	Medium	Medium	Incremental integration	Simplify integration
R07	Privacy/data exposure	Low	High	Medium	Synthetic/anonymized data	Remove affected data and reassess
R08	Poor usability	Medium	Medium	Medium	User testing	Refine interface
R09	Connectivity	Medium	Medium	Medium	Performance testing	Optimize

	limitations					MVP and document limitation
R10	Insufficient testing time	Medium	High	High	Prepare test cases early	Prioritize security/core-function tests

Risk management will be reviewed throughout the eight-week schedule rather than treated as a one-time planning activity.

6. Feasibility Checklist

Feasibility Area	Assessment	Rationale
Technical	Feasible	Laravel, PHP, relational databases, and web technologies are established technologies
Schedule	Conditionally feasible	Eight weeks is achievable only with strict MVP scope
Economic	Feasible	Primarily open-source and existing resources
Operational	Potentially feasible	Depends on stakeholder validation and organizational adoption
Security	Feasible with controls	RBAC, authentication, logging, and security testing are planned
Privacy	Feasible with controls	Synthetic/anonymized data reduces exposure
Usability	To be evaluated	Requires representative user testing
Connectivity	Risk requiring evaluation	Distributed operations may experience variable connectivity

Scope	Feasible	Core workplace functions are limited to a MVP
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7. Budget Estimate

ITEMS	DURATION	QTY	UNIT COST	TOTAL
I. - PERSONNEL	HOURS			
1.1 - Researcher/developer	200	1	CAD 30.00	CAD 6,000.00
1.2 - Academic supervision	20	1	CAD 60.00	CAD 1,200.00
1.3 - Sector Representatives	28	1	CAD 30.00	CAD 840.00
1.4 - St Luke Managers	15	1	CAD 30.00	CAD 450.00
1.5 - Test participants	20	1	CAD 20.00	CAD 400.00
1.6 - System Administrator	6	1	CAD 30.00	CAD 180.00
Sub-total :				CAD 9,070.00
II. - MATERIALS	MONTHS			
2.1 - Laravel/PHP		1	CAD -	CAD -
2.2 - Database software		1	CAD -	CAD -
2.3 - Git/GitHub		1	CAD -	CAD -
2.4 - ProjectLibre		1	CAD -	CAD -
2.5 - Browser/development tools		1	CAD 100.00	CAD 100.00
2.6 - Open-source security tools		1	CAD 100.00	CAD 100.00
2.7 - Existing development hardware		2	CAD 300.00	CAD 600.00
2.8 - Existing network/internet	2	1	CAD 50.00	CAD 100.00
2.9 - Synthetic/anonymized data		1	CAD -	CAD -
2.10 - Cloud/hosting contingency	3	1	CAD 100.00	CAD 300.00
2.11 - Backup/storage contingency		1	CAD 100.00	CAD 100.00
2.12 - Miscellaneous testing		1	CAD 200.00	CAD 200.00
Sub-total :				CAD 1,500.00
Total (Personnel + Materials)				CAD 10,570.00
Contingency (10% of the total)				CAD 1,057.00
GRAND TOTAL:				CAD 11,627.00

This estimate provides projected costs based on the Foundation's existing resources, the open-source softwares, and the identified risks.

Conclusion

The SL-DWP project presents a focused and feasible IT problem: the lack of an integrated, secure, role-based digital workplace for selected information-sharing and communication functions across St. Luke Foundation's distributed operations. The project addresses the problem through an eight-week MVP rather than attempting an unrealistic full enterprise deployment.

The proposal directly addresses the four requirements of the assignment by defining the problem and stakeholders, establishing a controlled scope and SMART objectives, identifying ethical and security implications, and documenting the project through lightweight planning tools. The Gantt chart, WBS, resource loading, budget, risk register, and feasibility checklist provide a traceable plan from requirements through evaluation. The resulting prototype will be assessed using functional, security, performance, and usability evidence, allowing the project to determine what is technically feasible rather than assuming organizational benefits in advance.

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